ASWIN PULICKAL BINDURAJ

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PROFESSIONAL SUMMARY

Full Stack Developer and M.Sc. student at the University of Koblenz, specializing in React.js, Django, and AI integration. Proven track record in building ML-driven applications and implementing Generative AI (LLMs) solutions. Expert in bridging complex backends with modern frontends and managing cloud deployments. Actively seeking a remote Working Student position to build impactful data products.

TECHNICAL SKILLS

- **Languages:** Python (Django, Flask), JavaScript (React.js, Node.js, Express, TypeScript)), SQL, HTML5, CSS3
- **Data & AI:** Machine Learning, NLP, LLMs, Pandas, NumPy, Scikit-learn, Matplotlib, TensorFlow
- **Data Analysis & Visualization:** Microsoft Excel, Google Sheets, Power BI, Matplotlib
- **DevOps & Cloud:** Docker, AWS (EC2), Git, CI/CD, TDD, Nginx, Celery, Redis
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB

LANGUAGES : English (C1 Level) | **German** (B1 Level) | **Malayalam (**Native)

EDUCATION

M.Sc. in E-Government (Web Science) | University of Koblenz, Germany | Oct 2024 – Present (Expected Sep 2027)

- Focus: Data Science, Network Analysis, BPM, Public Sector Digitization.
- **Level:** EQF Level 7.

Bachelor of Computer Applications (BCA) | Mahatma Gandhi University, Kottayam, India | 2020 – 2023

- Grade: **2.1 (German Equivalent)** / CCPA: **7.72/10** (Grade: **A**)
- Key Modules: Advanced Statistical Methods (**A+**), Operations Research (**A**)
- **Level:** EQF Level 6 (3-year Bachelor's).

CERTIFICATIONS: Certified Data Scientist (NCTT) & Python Full-Stack Developer (Luminar/LCC).

PROFESSIONAL EXPERIENCE

Backend Developer Intern | **Calvergy** – Aachen, Germany (Remote) *Dec 2025 – Feb 2026*

- Developed a high-performance Python engine for Heat Pump COP modeling using Carnot Efficiency and Pandas/NumPy, implementing automated data pipelines that converted raw sensor telemetry into standardized Kelvin-scale datasets for rigorous stress-testing and simulation.
- Architected RESTful APIs using Django and Docker under TDD principles, optimizing database schemas and query logic to reduce data retrieval latency by 20% while managing collaborative code deployments via Git/GitHub.

Data Science Specialist Intern | Luminar Technolab – Kochi, India *Mar 2024 - Nov 2024*

- Analyzed **15,000+ operational and customer records** using predictive modeling and sentiment analysis (up to **92% accuracy**) to identify **revenue trends and performance indicators.**
- Built clear **Power BI dashboards** to visualize analytical findings and support **data-driven decision-making**, alongside managing AWS cloud deployments.

Software Developer Intern | Soft Solutions – Kochi, India *Feb 2020 - Aug 2020*

(Note: Proactively completed consecutive full-time internships to transition into tech during COVID-19 university admission delays, before starting my BCA).

- Developed scalable full-stack web solutions using **Python,** supporting **500+ daily active users** and ensuring **99.9% database** uptime.
- Conducted rigorous software testing and debugging, achieving **95% code coverage** and **reducing production bugs by 30%.**

Full-stack Developer Intern | LCC Computer Education – Aluva, India *Jul 2019 - Jan 2020*

- Built responsive frontend interfaces using React.js and Django, **reducing page load times by 40%** through optimized component rendering.
- Developed and integrated RESTful APIs using Django REST Framework (DRF), enabling seamless data communication between the React.js frontend and PostgreSQL database.

KEY PROJECTS

EV Range Simulator | **Live Demo** | **GitHub**

- Developed a Full-stack ML application using **FastAPI** and **React** to predict electric vehicle range degradation.
- Integrated an **XGBoost** model utilizing real-time environmental telemetry and geospatial data for high-accuracy predictions.

AI Career Path Predictor | **Live Demo** | **GitHub**

- Developed a Full-stack application using **Django** and **React** that integrates **Google Gemini AI** to provide automated career stability analysis and personalized transition roadmaps.
- Engineered a predictive scoring engine with **Scikit-learn** and managed end-to-end deployment on **Render/Vercel**, including complex API configurations and environment management.

AI Document Intelligence Engine | [Live Demo] | [GitHub]

- Architected a Full-Stack **RAG (Retrieval-Augmented Generation) system** using **FastAPI** and **React**, implementing **LangChain** and **ChromaDB** to enable semantic search and precise information extraction from unstructured PDF data.
- Engineered a grounded **AI pipeline** utilizing **Cohere's Command-R** and **Rerank** models, featuring a custom "strict-context" prompt strategy to eliminate hallucinations and ensure 100% accuracy in document-based responses.

Koblenz,04.03.2026